Christley Olubela

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Education

University of Ibadan — **B.Sc. in Computer Science**

The International School, Olabisi Onabanjo University — **SSCE**

Technical Skills

- **Languages:** Rust, Python (Python 3.9–3.14), TypeScript, JavaScript, Solidity (0.8.x), SQL, C/C++, MQL5
- **Core Rust & Systems Engineering:** Trait-based generic API design, multi-crate Cargo workspaces, zero-copy deserialization, async/await (Tokio, Rayon), Write-Ahead Logging (WAL) with CRC32 recovery, CPython Stable ABI native extensions (PyO3, Maturin), doctests, property testing (proptest)
- **EVM Internals & MEV Execution:** REVM in-process simulation, Alloy (alloy-rs), Reth, op-geth IPC, Foundry (forge, cast, fuzzing), Flashbots Relay, Titan Builder, Beaver Build, bundle submission (`eth_sendBundle`, `eth_sendRawTransactionSync`), Arbitrum Timeboost, on-chain profit gates, Gnosis Safe, TimelockController
- **Solana Systems Infrastructure:** Yellowstone Geyser gRPC streaming, LiteSVM in-process simulation, Jito multi-region bundles, staked QUIC, Address Lookup Tables (ALT), compute-unit budgeting, Pyth pull oracles
- **DeFi Protocols & Liquidations:** Flash loans (Aave V3, Balancer V2, Morpho Blue, MakerDAO DssFlash, Kamino Lend, Solend), protocol liquidations across 6+ lending markets, cross-DEX atomic arbitrage, AMM curve mathematics (CPMM, StableSwap, Concentrated Liquidity, DLMM), ERC-4626 vaults
- **Backend, Databases & DevOps:** FastAPI, Axum, Actix, PostgreSQL (`psycopg_pool`), MySQL, Redis, SQLite, Docker, Docker Compose, Linux/systemd, Git, CI/CD (GitHub Actions), Prometheus, Grafana, bare-metal deployment
- **Open Source & Packaging (8,500+ Downloads Across 21 Packages):** Published maintainer across PyPI (7,000+ downloads) and crates.io (1,500+ downloads), comprehensive documentation sites (mdBook, docs.rs)
- **Certifications:** IBM Data Science Professional, IBM AI Engineering Professional, IBM AI Developer Professional

Professional Experience

Rust & Systems Engineer | Algorealm Inc. — **Jan 2026 – Mar 2026**

- Engineered high-throughput automated execution infrastructure in Rust and Python for delta-neutral perpetuals on Hyperliquid, managing sub-millisecond signal pipelines and dynamic risk limits.
- Built real-time market-data ingestion and mempool monitoring pipelines utilizing WebSockets, reducing order latency by 35% and optimizing fill timing under volatile market conditions.
- Implemented stochastic volatility forecasting modules to drive automated position sizing, dynamic stop-loss triggers, and automated capital protection ceilings.
- Developed quantitative modeling tools for structured liquidity exposures; computed payoff and impermanent loss curves as a function of price-path variance.

Rust & Systems Mentor | H.E.R. DAO Rust School — **Apr 2026 – Present**

- Mentored emerging software engineers in systems programming, memory safety models, and concurrency patterns for building resilient, high-availability distributed backend services.
- Grounded curriculum in production systems architecture, using the open-source crate Driftbrake to demonstrate trait design, semantic versioning discipline, and doctested API engineering.
- Demonstrated Rust for quantitative computing: safe concurrent computation, deterministic pricing execution, and performance-critical data pipelines.

Team Lead (Volunteer) | The Standing Church International — **Jun 2026 – Oct 2026**

- Led a team of 7 volunteers coordinating field operations and high-volume outreach reaching 200,000 individuals across 37 days; directed transit logistics across a 10-bus fleet for regional gatherings of 3,000+ attendees.
- Built centralized data tracking systems in Google Sheets to analyze contact feedback, optimizing follow-up scheduling around peak engagement windows and systematizing workflows that directly drove requirements for REACH.

Backend Engineer Intern | Beetle — **Mar 2025 – Jun 2025**

- Built automated execution backends and market-data streaming pipelines with MySQL persistence, ensuring deterministic execution and zero data loss under peak load.
- Engineered conversational query interfaces integrating OpenAI APIs with backend telemetry services, enabling engineering operators to query system states and risk metrics dynamically via natural language.

- Designed and deployed robust backend workflows ensuring high reliability and deterministic execution under live market volatility.

Flagship Open-Source Ecosystem (8,500+ Total Downloads Across 21 Packages)

Driftbrake | *Rust, Python (PyO3/ABI3), REVM, Tokio, WAL* **7 Crates** | **800+ Downloads** | **crates.io & PyPI**

- Published dual-ecosystem simulation and drift-detection library on PyPI (`driftbrake-py`) and crates.io (800+ total downloads) utilizing CPython Stable ABI native bindings and zero-dependency core traits.
- Designed a crash-resilient binary Write-Ahead Log (WAL) with IEEE 802.3 CRC32 verification, torn-write truncation recovery, and exclusive process locks.
- Implemented a 6-guard institutional halt hierarchy (Fast-Trip, Slow-Window Moving Average, Revert-Burst, Cumulative Revert Gas Budget, Volume-Weighted Average Realization, and High-Watermark Drawdown Caps) stopping execution when simulation-to-reality drift exceeds bounds.
- Shipped comprehensive documentation suite: mdBook architectural whitepaper, generated docs.rs reference, security policy, and doctest-verified quickstart.

Optimal-Sizing (v2.0.0) | *Rust, Cargo Workspace, PyPI, WASM* **10 Crates** | **700+ Downloads** | **crates.io & PyPI**

- Published 10-crate mathematical workspace on crates.io (`sizing-core`) and PyPI (`optimal-sizing`) establishing provable profit-maximizing trade sizes across Constant Product (CPMM), StableSwap, Velodrome Stable, and DLMM curves.
- Replaced naive grid searches with rigorous mathematical guarantee tiers: closed-form analytic roots (`ProvenOptimal`), Newton-Raphson solvers with proven convergence bounds (`NumericallyGuaranteed`), and golden-section search with active runtime unimodality checks.
- Engineered Dual Water-Filling capital allocation algorithm solving joint multi-pool budget distribution via Lagrange dual bisection in fixed-point decimal arithmetic with EIP-4844 rollup blob gas pricing.
- Built real-time WebSocket streaming daemon (`sizing-watch`) with `eth_subscribe("newHeads")` and historical multi-pool replay backtesting harness.

Brain-Libraries | *Python, PostgreSQL, AST Parsing, PyPI* **7 PyPI Packages** | **7,000+ Downloads**

- Architected and published a modular suite of 7 pip-installable Python packages on PyPI decoupling quantitative research pipelines with 86 automated test cases, AST algebraic deduplication, and PostgreSQL connection pooling via `psycopg_pool`.
- Engineered a 10-axis orthogonal transformation engine enforcing decorrelation constraints ($|\rho| < 0.70$) across high-dimensional feature spaces while preserving 85%–95% of predictive Sharpe ratio.
- Implemented tri-factor meta-synthesis combining Options surfaces, PEAD sentiment models, and systematic risk factor premia with Hierarchical Risk Parity (HRP) portfolio weighting.

MEV, Liquidation & High-Throughput Execution Systems

Corvus | *Rust, Solidity 0.8.24/Foundry, op-geth IPC* **$1.47M Modeled Net Profit** | **584 Events**

- Built high-performance two-chain liquidation engine covering Morpho Blue's $1.89B active debt on Base and Aave V3's WBTC reserve on Arbitrum; modeled $1.47M net annual profit ($122.8k/month) across 584 annual liquidation events ($90.01M volume).
- Implemented clip-size-aware DEX routing (Uniswap V3/V4, Camelot, Curve, Balancer) modeled from $50k to $1M clips with slippage parameters ranging 0.012% to 4.15%.
- Engineered on-chain profit gate contract (`FlashExecutor.sol`) enforcing atomic profitability, protocol whitelisting, and cold-wallet profit sweeps under immutable Gnosis Safe control; operated over local IPC in Frankfurt with sub-2.5ms sequencer ping.

Goshawk | *Rust, Solidity/Foundry, Reth IPC, revm* **$15.9B Active Borrowed Debt Covered**

- Engineered multi-protocol liquidation engine on Ethereum mainnet tracking $15.9B active debt across 6 protocols (Aave V3 $10.18B, Morpho Blue $2.19B, Spark $2.10B, Fluid $628M, Compound V3 $581M, Euler V2 $230M) against $27.2B supplied TVL.
- Modeled per-protocol gas break-even economics with local revm simulation and adaptive clip-sizing thresholds scaling dynamically during network congestion cascades ($38.50 to $280 per transaction gas).
- Formatted signed bundles for redundant low-latency submission to Flashbots Relay (88% builder market share), Titan Builder, and Beaver Build.

Gyrfalcon | *Rust 1.79+, Yellowstone Geyser gRPC, LiteSVM, Jito* **$367K Modeled Net Profit** | **613 Events**

- Developed sub-millisecond Solana liquidation engine targeting Kamino Lend ($1.03B debt), Save (Solend), and MarginFi; modeled $367k net annual profit across 613 annual events ($50.55M volume).
- Ingested streaming account state via Yellowstone Geyser gRPC into an in-memory lock-free obligation tracker; executed fee-free atomic flash borrowing via Kamino instruction introspection.
- Simulated every liquidation candidate in-process via LiteSVM before broadcasting Address Lookup Table (ALT)-compacted bundles across staked QUIC and multi-region Jito block engines (Amsterdam, New York, Tokyo, Frankfurt).

Skua | *Rust (async/Tokio), Solidity 0.8.24/Foundry, Docker Compose* **HyperEVM (Chain ID 999)**

- Engineered atomic flash-loan arbitrage and liquidation engine executing 4 strategies across HyperEVM DEXs, HyperLend, and HyperCore with on-chain profit validation callbacks.
- Designed multi-wallet nonce manager handling 8 nonces per address across 5–10 executor wallets, golden-section trade sizing, and synchronous `eth_sendRawTransactionSync` execution.

Kingfisher | *Rust (alloy-rs, Tokio, Rayon), Solidity/Foundry, systemd* **Arbitrum One**

- Built flash-loan arbitrage system capturing StableSwap price imbalances across imbalanced Curve pools, borrowing atomically from Aave V3 with zero capital risk via on-chain `require(netProfit >= minProfit)` guards.
- Exploited Arbitrum single-sequencer architecture via Timeboost ordering and co-located local Nitro node over IPC with sub-0.1ms block latency.

Kestrel | *Rust, Solidity/Foundry, Reth IPC, revm, SQLite* **Ethereum, Arbitrum, Gnosis**

- Monitored spreads between DEX pool prices and on-chain ERC-4626 redemption rates across yield vaults (sUSDS, sDAI, sUSDe, sxDAI), executing single-block atomic flash arbitrage with local revm pre-simulation.

Quantitative Infrastructure & Distributed Platforms

REACH Platform | *Python 3.11, FastAPI, React 18, TypeScript, PostgreSQL* **5-Tier RBAC** | **Real-Time Analytics**

- Architected enterprise field-operations platform featuring a 5-tier Role-Based Access Control hierarchy, invite-only token provisioning, and mobile-first data synchronization tracking thousands of field contacts in real time.
- Designed asynchronous REST APIs with FastAPI, database schema migrations with Alembic, and responsive React 18 dashboards visualizing regional outreach conversion metrics.

Osprey | *TypeScript, React/Vite, Hyperliquid SDK, FastAPI* **164 Passing Tests** | **20–100 Positions**

- Engineered automated quantitative portfolio system harvesting hourly funding rate yield on Hyperliquid perpetuals while systematically neutralizing directional market beta across 20–100 concurrent positions.
- Built automated market regime classifier (HOT, NEUTRAL, COLD) tuning hedge sizing and rebalance frequencies; verified by a 164-test suite with 5,000-cycle soak simulations.

HMBQT (Hummingbird Quant) | *TypeScript, Rust, Drift Protocol SDK, Helius RPC* **Ranger Finance Hackathon**

- Built order-flow trading vault on Solana (Ranger Earn) trading perpetual futures on Drift Protocol, ingesting real-time L2 orderbooks and fills via Drift DLOB WebSocket with Helius RPC execution.